Did F1's Cost Cap Level the Grid?

An Analysis of Competitive Balance, 2010-2025



B05 - Adrian Marcell, Ashwin Pillanda Ravindra, Jenniffer Hierro Cruz, and Saleha Hassan Alvi

BOSTON UNIVERSITY

Questrom School of Business

Formula 1 & The Cost Cap

Our Question: Did competition become more balanced after the 2021 cost cap implementation?

FIA Formula One World Championship Overview

How Formula 1 Works

- Teams compete in the Constructors' Championship.
- Teams accumulate points throughout the season.
- Championship standings rank teams by their total points.



What Changed in 2021

- Formula 1 introduced a cost cap.
- The cap limits specified team spending.
- Our analysis examines whether competition became more balanced afterward.

Sources: Formula 1, "The 2021 rules and regulation changes" (formula1.com).

Why Competitive Balance Matters

Why should fans, teams, and sponsors care about a more competitive grid?

FANS

Closer racing can make each race more exciting and keep fans engaged.

SPONSORS

More teams competing near the front can create additional sponsorship opportunities.

F1 & TEAMS

A more balanced grid can help smaller teams compete and strengthen F1's appeal.

Our focus: Did the cost-cap era spread competitive success across more teams?



Sources: Liberty Media Investor Presentations, FIA Financial Audits, Team B05 Analysis.

Data & Measures

How did we clean our dataset and define our measures?



Data Measurement Overview



Data

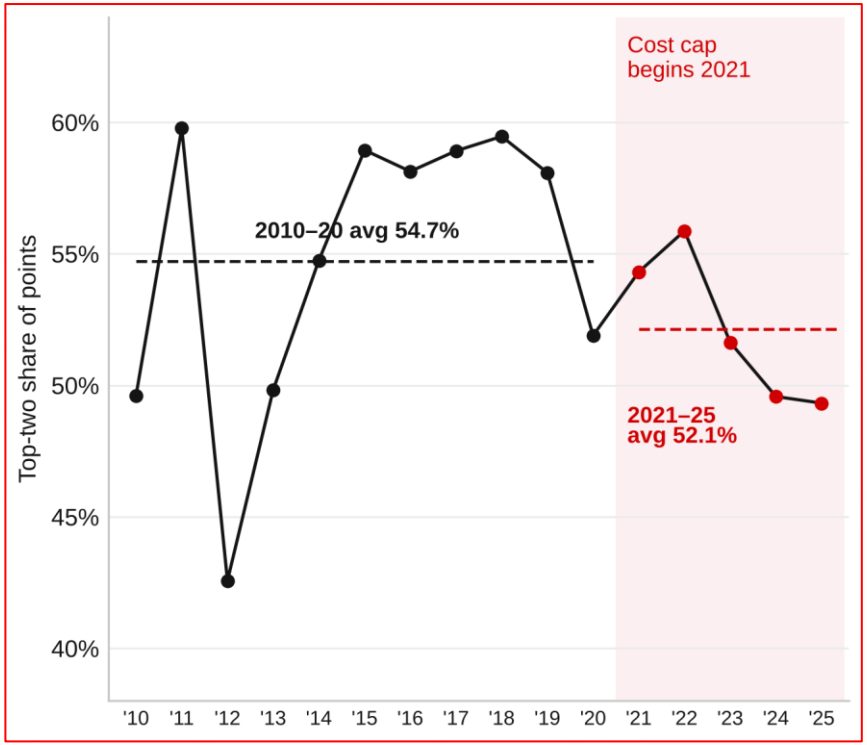
- **Source:** F1DB constructor standings and Grand Prix results.
- **Pre-cap:** 2010 - 2020 (11 seasons).
- **Post-cap:** 2021-2025 (5 Seasons).
- Renamed teams grouped consistently.

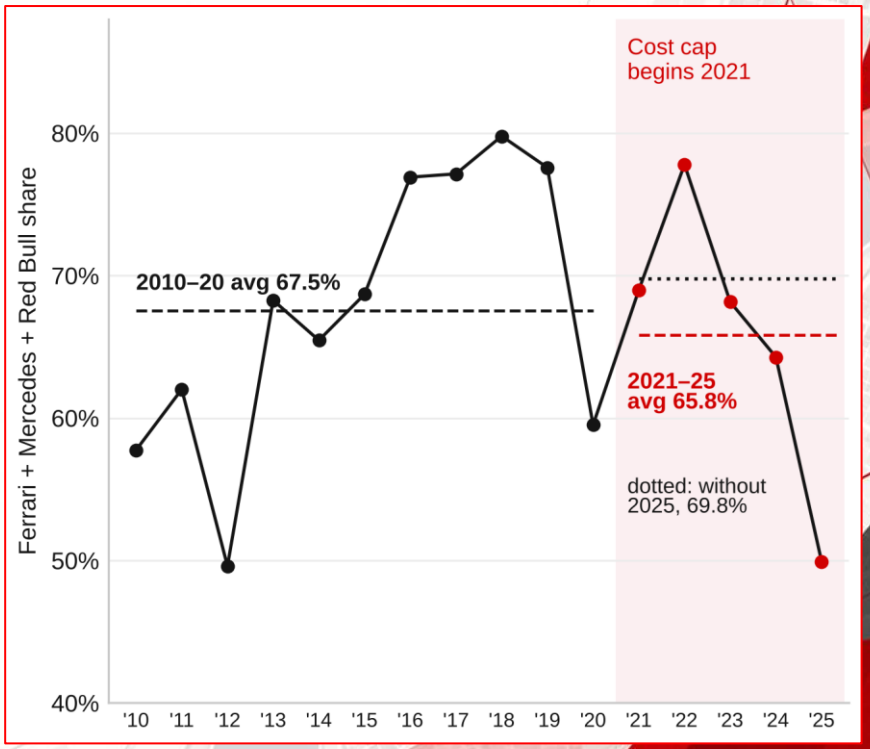
Measures

- **Point share:** (team point / all team points in season) x 100.
- **Top-two share (CR2):** combined share of P1 and P2 .
- **Midfield Share:** combined share of P4 through P7.
- **Big Three Share:** combined points share of Ferrari, Mercedes, and Red Bull.

Sources: F1DB contributors, release v2026.15.1 (github.com/f1db/f1db), CC BY 4.0; Formula 1, “Racing Point Force India entry” (formula1.com).

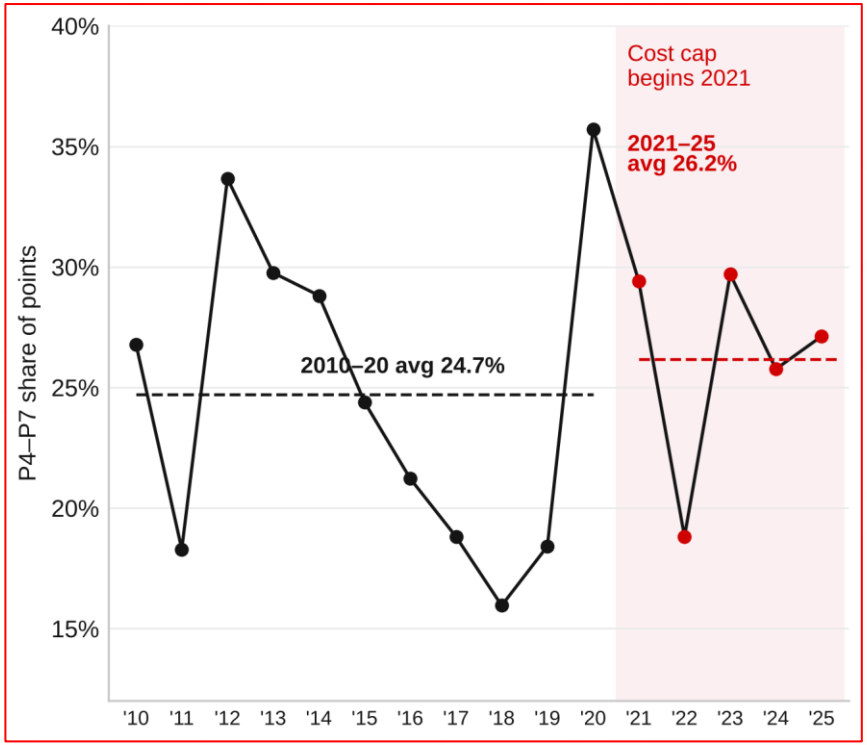
Top-team Concentration Fell Slightly

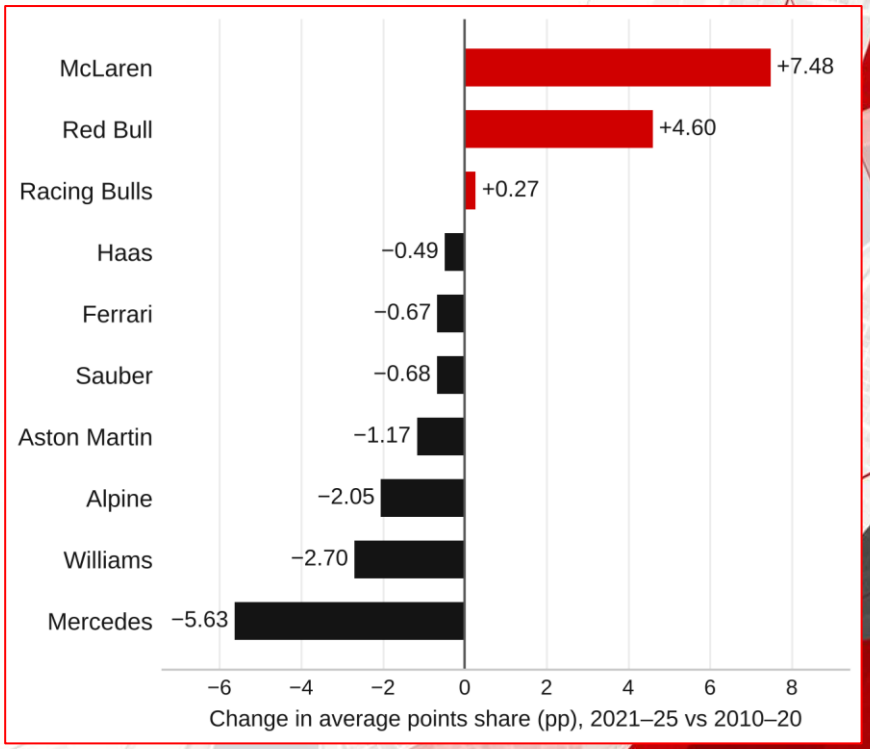




Sources: F1DB contributors, release v2026.15.1 (github.com/f1db/f1db), CC BY 4.0, filtered and aggregated by the team; SciPy developers, scipy.stats.ttest_ind (Welch's test).

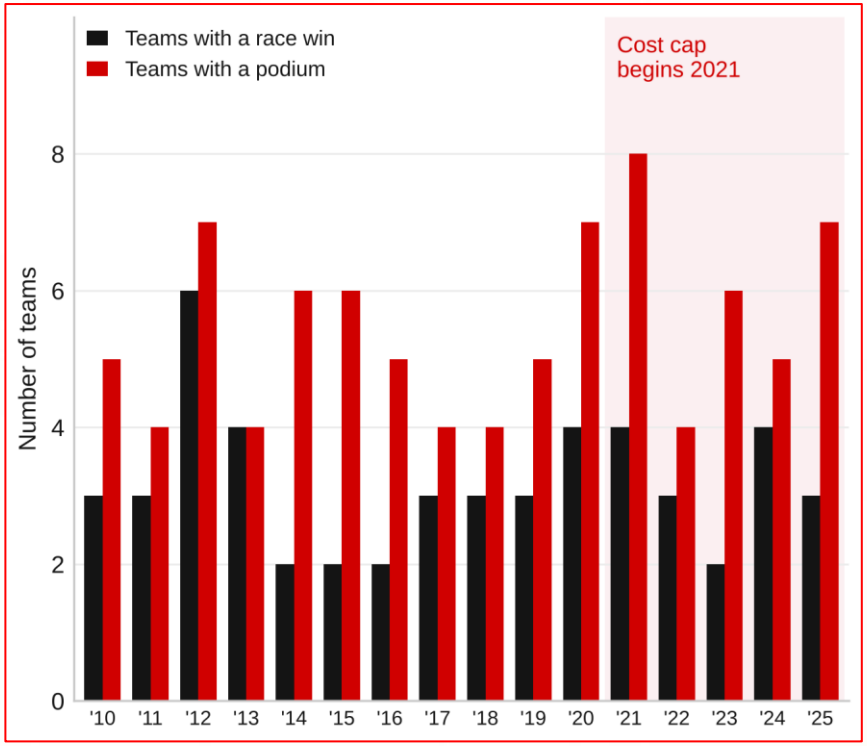
Gains were Concentrated in a Few Teams

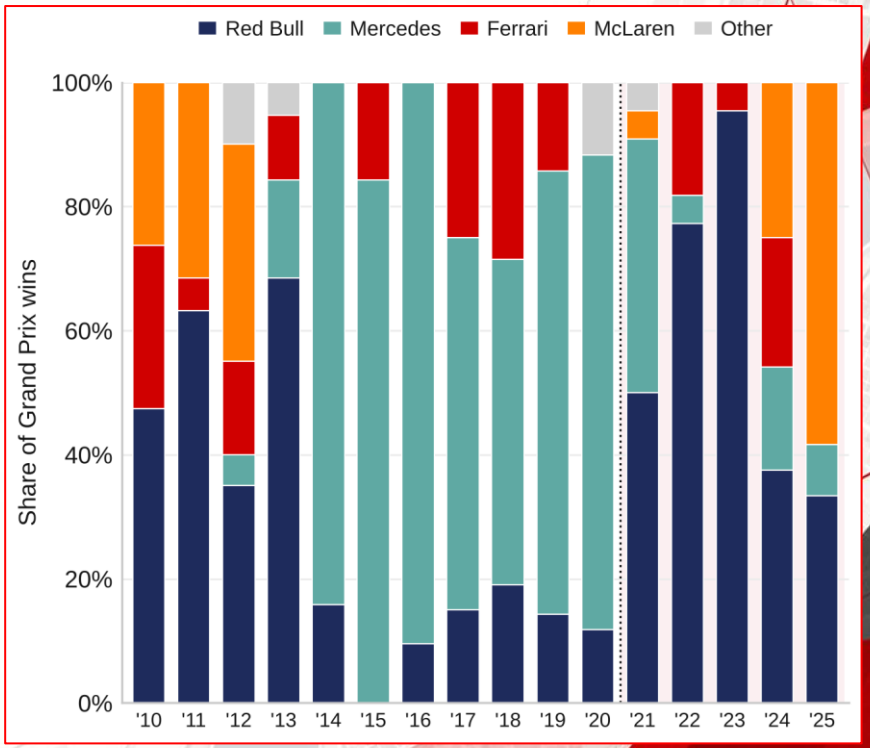




Sources: F1DB contributors, release v2026.15.1 (github.com/f1db/f1db), CC BY 4.0; Formula 1, "McLaren year by year" (formula1.com), for McLaren's 2024 and 2025 constructors' titles.

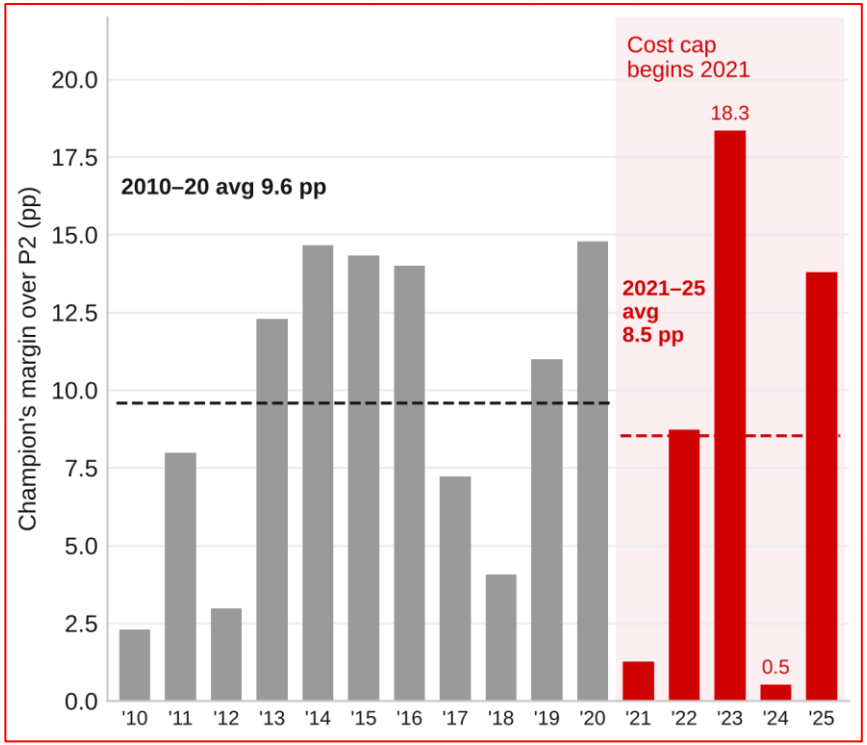
More Teams Reached Podium, Wins Remain Concentrated

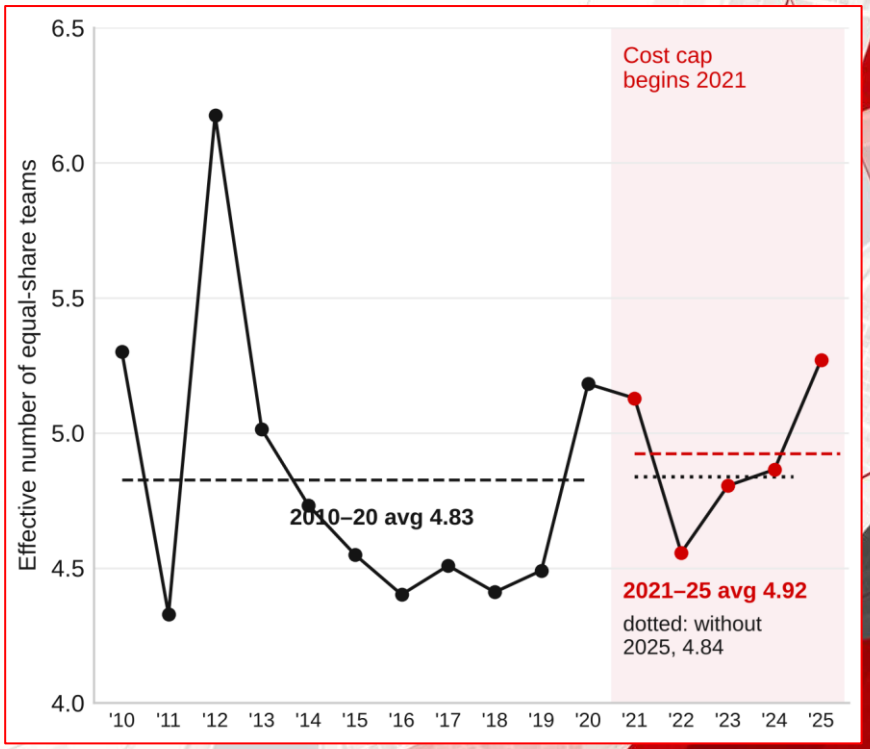




Sources: F1DB contributors, release v2026.15.1 (github.com/f1db/f1db), CC BY 4.0: Grand Prix results and constructors chronology, aggregated by the team.

Close Title Fights didn't Consistently Extend Across Fields





Sources: F1DB contributors, release v2026.15.1 (github.com/f1db/f1db), CC BY 4.0; U.S. Department of Justice, "Herfindahl–Hirschman Index" (justice.gov/atr), for the HHI definition.

What Should Stakeholders Do?

Practical steps to drive results, supported by our findings







Stakeholder	Recommended Action	Evidence & Limitation
FIA	Require a competitive-balance review across a full rules cycle before changing financial rules. Include both baselines and results excluding the latest season.	Conclusions change with the comparison period. Points data cannot isolate the cost cap's effect.
Middle-Ranked Teams	Set performance targets across several seasons and benchmark their progress against improving teams.	McLaren gained considerably, while several midfield teams lost share. The analysis does not reveal their spending decisions.
Sponsors	Include emerging contenders in sponsorship shortlists and use performance conditions in deals.	Podium opportunities expanded outside the Big Three, particularly in recent seasons. Sponsorship returns were not measured.

Sources: F1DB contributors, release v2026.15.1 (github.com/f1db/f1db), CC BY 4.0; Formula 1, "The 2021 rules and regulation changes" and "Seven key rule changes for 2022" (formula1.com).

Conclusion

Small shifts in competitive balance, not a clear transformation

What We Found

- The top two teams earned a slightly smaller share of points **−2.58 pp.** (p = 0.245)
- Midfield teams gained some ground, but the increase was small. **69.8%** Big Three share without 2025 (vs 65.8%)
- The number of teams winning races barely changed. (**3.18 → 3.20)**



What This Means

- Different teams had success, but the grid is still dominated by a few.
- We cannot say the cost cap caused these changes.
- We need more seasons to see if the pattern continues.



Sources: F1DB contributors, release v2026.15.1 (github.com/f1db/f1db), CC BY 4.0; Formula 1, 2021 and 2022 rule-change articles (formula1.com); SciPy developers, scipy.stats.ttest_ind.